

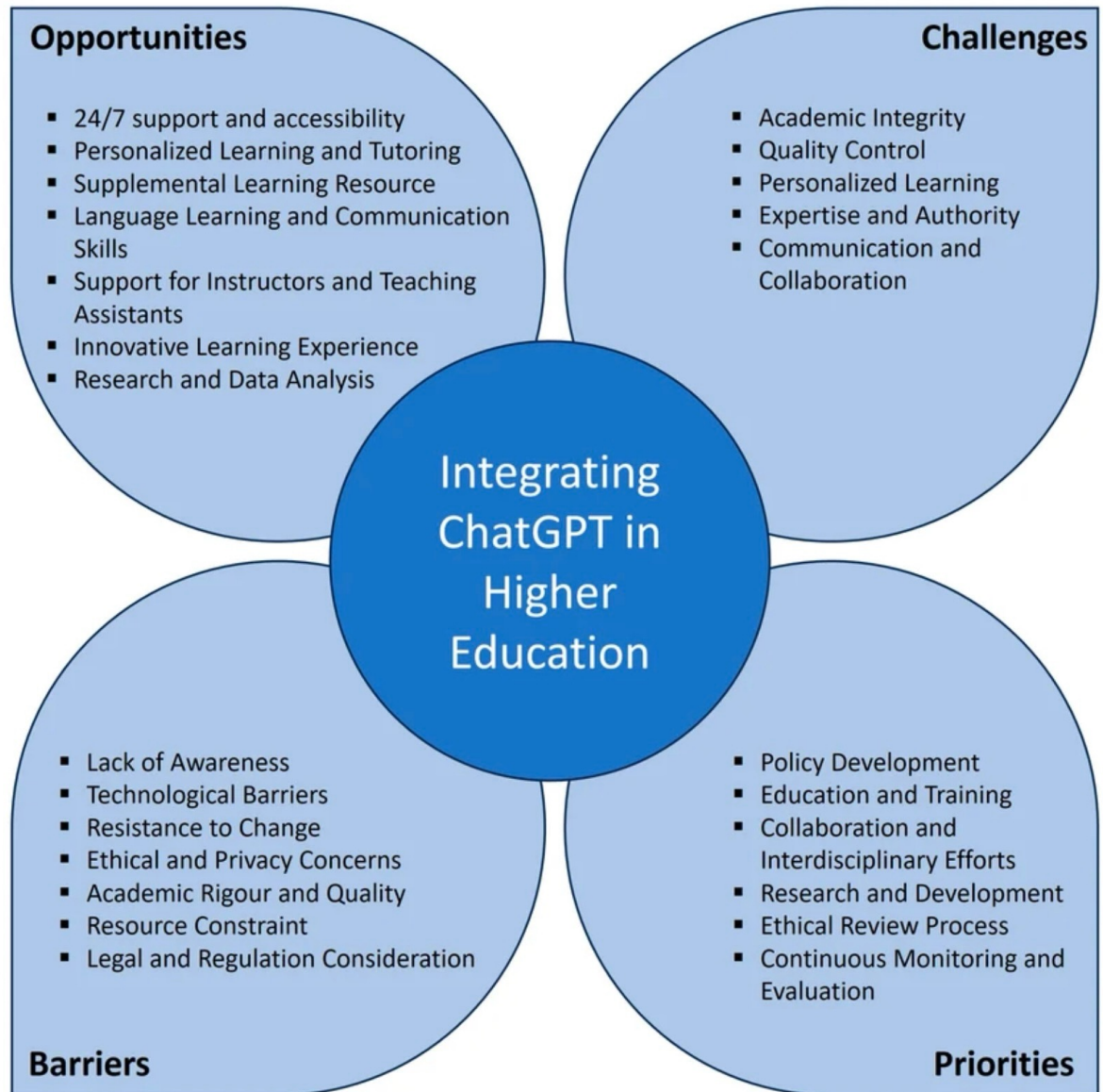
Academic Integrity and Responsible Use of AI

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Michel-Villarreal, Rosario & Vilalta-Perdomo, Eliseo & Salinas Navarro, David Ernesto & Aguilera, Ricardo & Gerardou, Flor. (2023). Challenges and Opportunities of Generative AI for Higher Education as Explained by ChatGPT. Education Sciences. 13. 856. 10.3390/educsci13090856. <https://www.researchgate.net/publication/373328547/figure/fig1/AS:11431281420312198@1746232951628/Themes-emerging-from-the-interview-own-elaboration.tif>





Topics

Misuse of AI and Academic Integrity: plagiarism and AI Policy, over-reliance, deskilling

Responsible use of AI: common principles and ethical guidelines in academic integrity and AI support for learning.

Critique current academic misconduct policies regarding AI to determine if they adequately address the nuances of co-authorship and automated generation.

Write a policy for using AI in an educational context.

Misuse of AI and Academic Integrity





Why is Academic Integrity Important?

- Academic Integrity is crucial to the learning process.
- Violation of Academic Integrity (i.e., academic misconduct) gives an unfair advantage to those who take dishonest shortcuts.
- Honest work builds knowledge, skills, and self-esteem.
- Cheating undermines the quality of education for everyone, and thus reduces the real value of a university education.
- Cheating harms the reputation of the University and diminishes the credibility of a degree from the University of Sydney.

Plagiarism and the use of AI

Plagiarism is the act of presenting someone else's work, ideas, or intellectual property as one's own without proper acknowledgment or permission. It involves copying or closely imitating another's work and claiming it as original.

Scenario 1: AI-Generated Content Without Attribution A student uses an AI tool to generate a significant portion of the course report—such as entire paragraphs, analyses, or conclusions—and submits this work as their own without citing the AI or specifying its contribution. This could be seen as plagiarism because the student is presenting AI-generated work as their original effort, without acknowledgment.

Scenario 2: AI as a Writing Aid with Proper Attribution A student uses AI to assist with brainstorming ideas, checking grammar, or generating outlines. The student writes the report independently, incorporating their own analysis and understanding. If the student acknowledges that they used AI tools as part of their process, and the final work reflects their own input, this would not be considered plagiarism.



Ambiguous scenario

A student uses an AI tool to generate a summary of research articles related to their report topic. The student then paraphrases parts of the AI-generated summaries, incorporates some of their own interpretation, and cites the original articles directly in their report. They submit the report, stating they used AI for "assistance with summarizing sources."

Ambiguity:

Is the student's use of AI acceptable because they provided proper citations and added personal analysis, or does it constitute plagiarism because they relied heavily on AI-generated content without explicit acknowledgment of the AI's role?

This scenario blurs the line between responsible use of AI as a research aid and potential intellectual dishonesty, depending on institutional policies and expectations regarding AI transparency.



University of Sydney AI Policy

- 2024: <https://www.sydney.edu.au/news-opinion/news/2024/11/27/university-of-sydney-ai-assessment-policy.html>
- 2025: <https://www.sydney.edu.au/students/academic-integrity/artificial-intelligence.html>

From Semester 2, 2025, the University has implemented a 'two-lane' approach to assessments.

- **Lane 1: Secure assessments**

These in-person, supervised assessments are used to ensure students have acquired the skills and knowledge required in their unit. students are generally *not* permitted to use AI in **secure (supervised)** tasks, such as exams and tests, unless the Unit of Study coordinator has given express permission in the Unit Outline. Using AI when not allowed could amount to a breach of academic integrity for which you could be investigated.

- **Lane 2: Open assessments**

These unsupervised assessments support the development of disciplinary knowledge and skills. They allow students to engage productively and responsibly with AI tools as part of their learning experience. For **open (unsupervised)** assessments, students *will* be able to use AI, and need to appropriately acknowledge its use – provided they do so, this would not be a breach of academic integrity.



Misusing AI: Academic Integrity at the University of Sydney

Misusing generative AI can breach the Academic Integrity Policy.
Examples include:

- Using generative AI such as chatbots, paraphrasing tools or translation tools in an assessment where their use has been prohibited.
- Submitting AI-generated work without appropriate acknowledgment.
- Inputting University teaching or course materials, content generated by another student, Intellectual Property from external partners, or any person's personal or health information into AI tools.



Consequences of Academic Integrity Breaches

- Minor Breaches (e.g., poor referencing): Often handled by unit coordinators, involving a mark reduction or requirement to complete academic honesty training.
- Serious/Major Breaches (e.g., plagiarism, cheating, collusion): Can lead to a zero grade for the assessment, failing the unit of study, or being excluded from the university.
- Persistent Breaches: A history of breaches may lead to more severe penalties.
- Suspension/Expulsion: Severe cases can result in temporary or permanent removal from the university.
- Visa/Legal Risks: International students risk visa cancellation, and contract cheating can lead to blackmail.



University of Sydney Policy: Acknowledging the use of AI

If you are permitted to use automated writing tools or generative AI by your unit coordinator, you will need to include a statement in your submitted work explaining:

- that you have used AI tools in completing the work
- the name and version of the automated writing or generative AI tools you have used
- the publisher
- the uniform resource locator (URL)
- and a brief description of how you have used them.
- If your unit coordinator has made any additional stipulations about how they want the class to acknowledge AI use, you should also follow these rules.



More comprehensive acknowledgement

When permitted to use AI tools, students **must include** in their submission:

- A clear **acknowledgment statement** detailing:
 - AI tool names and versions.
 - Publishers and URLs.
 - How the AI was used (e.g., for drafting, editing, brainstorming).
- An **AI interaction transcript or appendix**, demonstrating the student's role and engagement:
 - Sample prompts entered.
 - Responses received.
 - Description of how these exchanges influenced the final work.

Justification:

Ensures transparency, holds students accountable, and provides verifiable evidence of ethical AI engagement.



Responsible Use of AI in Education





Responsible Use of AI in Higher Education

Principles

- Academic Integrity
- Disclosure
- Data Privacy
- Bias Awareness
- Human Oversight
- Informed Consent
- Continuous Monitoring



Data Privacy and the responsible use of AI in Higher Education



Data privacy refers to the safeguarding of personal, sensitive, or identifiable information from unauthorized access, use, disclosure, disruption, modification, or destruction. In the context of AI in higher education, data privacy is a critical component to ensure ethical and responsible engagement with AI tools.

Responsibilities of students and universities

Students: Must be cautious about the information they input into AI systems, avoiding entering sensitive or personally identifiable data unless explicitly permitted and protected.

Universities: Are responsible for selecting AI tools with clear privacy policies, providing training on data privacy best practices, and establishing guidelines for ethical AI use.



Policy Considerations

Third-party AI Platforms: Many AI tools operate on external servers, raising concerns about data residency, access control, and compliance.

Data Breaches: Institutions must have protocols to identify, respond to, and mitigate potential data breaches involving AI systems.

Balancing Privacy and Utility: Ensuring privacy while maintaining the effectiveness of AI applications can be complex; strategies like data anonymisation are essential.



Why Bias Awareness Matters



In an academic setting, the responsible use of AI requires students and educators to recognise that AI tools may not be neutral or infallible. Uncritical reliance on biased AI outputs can compromise the integrity of research, assessments, or learning processes, and may propagate stereotypes or misinformation.

Key Elements of Bias Awareness

- **Critical Evaluation of AI Output:**

Students should question the suggestions, summaries, or analyses provided by AI tools rather than accepting them at face value. This includes being alert to possible biased language, stereotypes, or inaccuracies that may influence interpretation or decision-making.

- **Understanding Data and Model Limitations:**

Students should be aware that AI models are based on training data that may be incomplete or unrepresentative, affecting the fairness and accuracy of results. Recognizing these limitations helps in making informed judgments about AI-generated content.

- **Promoting Equity and Fairness:**

When using AI for research or assessment, students should consider whether the AI's outputs reflect diverse perspectives and avoid reinforcing inequalities. If biases are identified, students should actively work to mitigate their impact, such as by supplementing AI insights with diverse viewpoints or corroborating findings with multiple sources.

- **Ethical Responsibility:**

Aware students understand their ethical obligation to prevent the misuse or misrepresentation of biased AI outputs. This includes avoiding discrimination, ensuring neutrality, and advocating for equitable practices within their academic work.



Policy Considerations related to Bias Awareness

- **Bias Audits:** Regularly evaluate AI models for biases, especially concerning protected characteristics such as race, gender, or socioeconomic status.
- **Inclusive Datasets:** Use diverse and representative datasets to train AI systems, minimizing the risk of biased outputs.
- **Transparency and Explainability:** Promote explainable AI models that allow users to understand decision-making processes and identify potential biases.
- **Training and Awareness:** Educate staff and students on recognizing, understanding, and addressing biases in AI systems.



Risks of Over Reliance on Generative AI



Risks of Over Reliance on Generative AI

- Algorithmic Bias
- Impaired Creativity
- Reduced Critical Thinking
- Lack of Interpretation

<https://www.enago.com/academy/responsible-use-of-generative-ai/>

Risks of Overreliance on Generative AI

Algorithmic Bias

- AI algorithms trained on biased datasets may amplify existing societal prejudices if researchers solely rely on AI-generated insights.
- Overreliance on generative AI can limit critical examination of biases, leading to a restricted understanding of complex issues and alternative viewpoints.

Impaired Creativity

- Generative AI lacks human imagination and originality, hindering their ability to propose groundbreaking ideas or explore new areas.
- Sole reliance on AI-generated insights may suppress researchers' creativity and prevent the exploration of unconventional approaches in academic research and writing.

Reduced Critical Engagement

- Overreliance on generative AI can lead to reduced active engagement and critical thinking by researchers.
- Declined emphasis on critical thinking skills may compromise the depth and rigor of academic research and writing.

Lack of Interpretation

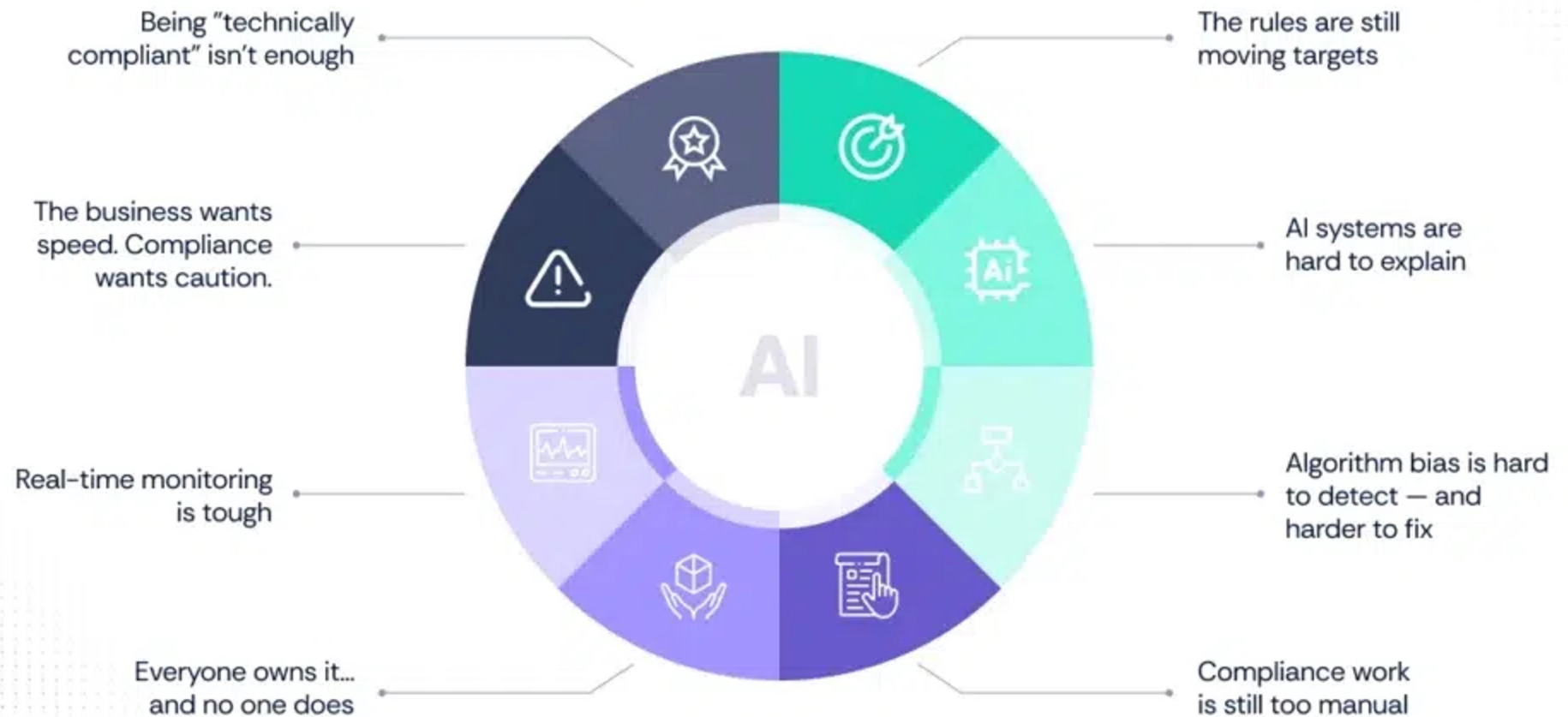
- AI algorithms can find it difficult to interpret and explain complex discipline-specific jargons.
- The black-box nature of AI algorithms raises questions about the trustworthiness and reliability of AI-generated results, requiring cautious assessment by researchers.

Scenarios and Principles



There are many policies and principles for the use of AI in education. Due to the ambiguity of specific cases and the rapid advances in AI technology, there needs to be ongoing discussions of the use, principles, and policy of AI in education

8 major challenges in becoming AI compliant



Principles for the Responsible Use of Generative AI in Homework Assignments

- **Original Thinking and Input:** Begin each assignment by independently attempting to solve problems, answer questions, or complete tasks. Only use generative AI after making a genuine effort to generate original ideas and solutions.
- **Cite AI-Generated Content Appropriately:** If you use generative AI to assist in your assignment, clearly identify and cite the AI-generated content within your work. Give appropriate credit to the AI tool used and acknowledge its contribution to your assignment.
- **Integrate AI-Generated Content Ethically:** Integrate AI-generated content thoughtfully and purposefully into your assignment, ensuring that it aligns with the context and topic of the task and enhances your understanding rather than overpowering your original input.



More Principles for Responsible Use

- **Review and Edit AI-Generated Content:** Thoroughly review and edit any content generated by AI to ensure accuracy, relevance, and coherence with the assignment requirements. Personalize the content to match your style and approach.
- **Respect Academic Honesty Policies:** Adhere to the academic honesty policies set by the university and instructors. Refrain from using generative AI to create or copy content that violates these policies, including plagiarism or unauthorized collaboration.
- **Ask for Clarification from Instructors:** Seek guidance from instructors if you are unsure about whether the use of generative AI is appropriate for a specific assignment or how to use it in an ethical and responsible manner.
- **Be Transparent about AI Usage:** If the assignment allows, disclose the use of generative AI to your instructors, providing them with insights into how you utilized it to complete the assignment responsibly.



Scenarios where AI usage Can Pose Challenges to Academic Integrity

- **Plagiarism:** If AI-generated content is used without proper citation or acknowledgment, presenting it as one's own work constitutes plagiarism. Copying or submitting AI-generated text without appropriate attribution violates academic honesty.
- **Misrepresentation of Knowledge and Skills:** Relying solely on AI-generated responses to assignments can misrepresent a student's true understanding of the subject matter, leading to inflated grades and inaccurate perceptions of academic capabilities.
- **Fabrication of Results or Data:** AI tools that generate data, experimental results, or findings can be misused to fabricate research outcomes, leading to false conclusions and compromising the integrity of academic research.



More Scenarios

- **Violating Assignment Instructions:** If an assignment explicitly prohibits the use of AI-generated content and a student uses such content anyway, it violates the instructor's guidelines, affecting academic honesty and integrity.
- **Inappropriate Use of AI for Exam Purposes:** Using AI during exams or assessments to assist in answering questions, whether directly or indirectly, is a violation of academic integrity and exam regulations.
- **Evading Learning and Understanding:** Over-reliance on AI to complete assignments can hinder the development of critical thinking, problem-solving skills, and a deeper understanding of the course material.





Case Study: Irresponsible Use of AI in a Homework Assignment

Background: Alex is an undergraduate student enrolled in a computer science course that involves a programming assignment. The assignment requires students to develop a codebase for a specified problem using Python. The professor encourages students to seek guidance and use online resources, but emphasizes that the code should be written independently to assess each student's understanding of the material.

Scenario: Alex finds a powerful code-generating AI tool online that can create Python code for the given problem statement. Instead of using it as a learning aid, Alex uses the AI tool to generate the entire codebase for the assignment. They don't review the generated code or attempt to understand it. They submit the AI-generated code as their own work without disclosing or citing the use of the AI tool.

Why is this a case of irresponsible use of AI?

- **Lack of Originality:** John's submission lacked originality, and he did not demonstrate his understanding of the programming concepts required for the assignment. The AI-generated code did not reflect his own thinking or problem-solving skills.
- **Plagiarism:** By submitting the AI-generated code as his own without proper attribution or acknowledgment of the AI tool, John engaged in plagiarism, a clear violation of academic integrity.
- **Misrepresentation of Skills:** John's submission falsely represented his programming skills and knowledge to the professor. The AI tool was essentially doing the work for him, leading to a misleading assessment of his abilities.
- **Failure to Follow Assignment Instructions:** The professor had explicitly instructed students to write the code independently. John's use of the AI tool disregarded these instructions, violating the guidelines provided for the assignment.
- **Lack of Transparency:** John did not disclose his use of the AI tool to the instructor, failing to maintain transparency regarding the source of the code.





Case Study: Responsible Use of AI in a Homework Assignment

Background: Dru, an undergraduate student majoring in natural language processing, is tasked with a challenging language analysis assignment for their linguistics course. The assignment involves analyzing a complex text using linguistic principles and techniques.

Scenario: Dru uses an AI-powered natural language processing tool to assist in the analysis of the text. They input the text into the AI tool and carefully reviews the generated analysis, paying close attention to the linguistic patterns, semantics, and syntactic structures identified by the AI. Dru then incorporates the AI-generated insights into their assignment, citing the AI tool as a reference.

Why is this a case of responsible use of AI?

- **Understanding and Learning:** Dru used the AI tool to enhance their understanding of linguistic analysis. They actively engaged with the AI-generated analysis, allowing Them to grasp the concepts better and gain insights into linguistic patterns.
- **Supplementary Tool:** The AI tool served as a supplementary tool to aid Dru in their analysis. It was used to assist and guide them, not to replace their critical thinking or analysis skills.
- **Integration and Review:** Dru carefully reviewed the AI-generated analysis and integrated it thoughtfully into their assignment. They assessed the relevance and accuracy of the insights provided by the AI tool before incorporating them.
- **Proper Citation:** Dru cited the AI tool in their assignment, acknowledging its contribution to the analysis. This citation provided transparency about the tool's usage and its impact on their understanding and work.





Case Study: Ambiguous Use of AI in a Homework Assignment

Background: Deng is an undergraduate student majoring in mathematics and is assigned a complex problem set in calculus. The assignment requires solving intricate mathematical equations and deriving theorems related to advanced calculus.

Scenario: Deng uses an AI-powered mathematics solver tool to assist in solving the equations and deriving the theorems required for the assignment. The tool successfully provides solutions and derivations for the problems. Deng reviews the solutions, ensuring they align with the principles of calculus, and then integrates the AI-generated solutions into the assignment.

Why is this case ambiguous?

- **Level of AI Assistance:** The AI tool significantly aided Deng in solving complex mathematical problems. The debate arises from whether the extent of assistance provided by the AI tool was appropriate or if it crossed the boundary of acceptable aid.
- **Understanding vs. Tool Reliance:** Some argue that using the AI tool may undermine the student's understanding of the subject matter, while others contend that it can help students grasp complex concepts by providing correct solutions and guidance.
- **Integration of AI Solutions:** Deng reviewed and validated the AI-generated solutions to ensure correctness and alignment with calculus principles. The debate centers on whether this validation was sufficient to make the AI-generated content an acceptable part of the assignment.



Potential Outcomes

- If the professor views the AI tool as an acceptable aid and values the accuracy of the solutions rather than the method of arrival, Deng may receive a positive evaluation for the assignment.
- If the professor believes that the AI tool excessively assisted Deng, undermining the purpose of the assignment to test understanding, the use of the AI-generated content may be deemed unacceptable, potentially resulting in a lower grade or academic integrity concerns.





Case Study: AI-Generated Study Game for Exam Preparation

Background: Taylor is an undergraduate student majoring in computer science. They are preparing for a challenging programming exam that covers various topics. They decide to use an AI tool to generate a customized study game to help reinforce their understanding of the exam material.

Scenario: Taylor inputs specific topics and key concepts relevant to the exam into a specialized AI tool. They prompt the AI to create a study game that includes questions, quizzes, and interactive challenges related to the chosen topics. The AI tool generates a comprehensive game that allows Taylor to test their knowledge, identify areas of weakness, and improve their understanding of the subject matter.

Are there ethical issues in this use of AI?

Customized Study Tool: Taylor utilized AI to create a customized study tool tailored to the exam topics. While this demonstrates resourcefulness and innovation, it raises questions about whether other students have access to similar AI tools and whether it gives Taylor an advantage over those who do not.

Potential Unfair Advantage: If other students in the class are unaware of or lack the skills to utilize AI tools for creating customized study aids, Taylor may have an unfair advantage in exam preparation. The study game gives him an interactive and targeted tool that aligns closely with the exam content, potentially boosting his performance.

Access to AI Technology: The disparity in knowledge and access to AI tools can create an imbalance, where students like Taylor who are familiar with AI and its applications have an advantage over those who are not. This highlights the importance of equal access and awareness of AI tools among students.



Potential Outcomes

- If the professor is aware of Taylor's approach and deems it an acceptable use of available technology to aid in exam preparation, Taylor may benefit from improved understanding and performance on the exam.
- If the professor is concerned about potential unfair advantage due to varying levels of AI tool usage among students, they may consider adjusting the exam format or providing additional resources to level the playing field.



How to use generative AI in this unit

1 Human-First Attempt

- ✓ Spend at least 15 minutes attempting the problem without AI
- ✓ Identify the key concepts you are expected to understand
- ✓ Use AI for hints and guidance rather than the final answer.

2 AI as a learning partner

- ✓ Ask the AI to explain its reasoning, e.g. "Can you explain the logic behind this step"
- ✓ Test the AI with alternative answers or misconceptions
- ✓ Refine the AI using course-specific context

3 Final Check

- ✓ Test if you can explain the solution without AI
- ✓ Check the AI's response for mistakes or errors in logic
- ✓ Iterate on the prompt instead of accepting the first answer





Tutorial Activities

Critique the use of AI while writing a report for a course assignment.

Critique the use of AI while engaging in a technical assignment like writing code.

Critique the University of Sydney AI policy using the principles of the responsible use of AI in higher education.

Write a new AI policy for the University of Sydney.

Write an AI Policy for this unit of study



Quick Weekly Survey



ENGG112 week 2